

# HomeSenseKD: Heterogeneous Federated Knowledge Distillation for CSI-Based Indoor Sensing

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HomeSenseKD Research

<https://github.com/gadm21/HomeSenseKD>

**Abstract**—Indoor human activity recognition (HAR) using Wi-Fi Channel State Information (CSI) offers a non-intrusive, privacy-preserving alternative to camera-based sensing. Deploying such systems across multiple homes introduces three compounding challenges: statistical heterogeneity of activity distributions, architectural diversity of edge devices, and the prohibition on sharing raw sensor data. We present HomeSenseKD, a benchmark framework for heterogeneous federated knowledge distillation applied to CSI-based indoor sensing. Seven algorithms—FedMD, FedAKD, FedMKS, FedAvg, FedProx, Local, and Centralised—are evaluated across four datasets (HomeOccupancy, HomeHAR, MNIST, CIFAR-10) under both IID and non-IID partitions in two structured experimental phases. A dual-head tiered model decouples local classification from soft-label distillation, enabling model-heterogeneous collaboration without exchanging raw data or full model weights. On HomeOccupancy, FedMKS achieves 90.3% validation accuracy under IID partitioning, within 2% of the centralised ceiling. HomeHAR reveals the failure mode of federated learning under severe data scarcity. Code and datasets are publicly available.

## I. INTRODUCTION

Wi-Fi Channel State Information (CSI) encodes fine-grained multipath propagation effects that shift measurably when a person moves, breathes, or changes posture. By analysing CSI time series from commodity access points, it is possible to recognise indoor activities without deploying cameras or wearables—an increasingly attractive proposition for smart homes and assisted living [1], [2].

Training accurate CSI-based models, however, requires labelled data that is inherently private and difficult to collect at scale. Federated Learning (FL) mitigates this by allowing devices to collaborate without sharing raw data; each client trains locally and contributes only model updates or soft predictions [15]. Beyond privacy, FL brings a natural benefit for sensing: a client whose environment lacks certain classes (e.g., no “smoking” behaviour observed) can still learn to recognise them through knowledge transferred from peers that have seen those classes, as illustrated in Fig. 1.

A persistent practical obstacle is *model heterogeneity*: edge devices range from resource-constrained sensors to capable gateways, making a one-size-fits-all model architecture infeasible. Standard weight-sharing algorithms (FedAvg, FedProx) require all clients to share the same architecture; knowledge-distillation approaches (FedMD, FedAKD) avoid weight exchange entirely but forego the benefits of model aggregation. FedMKS bridges

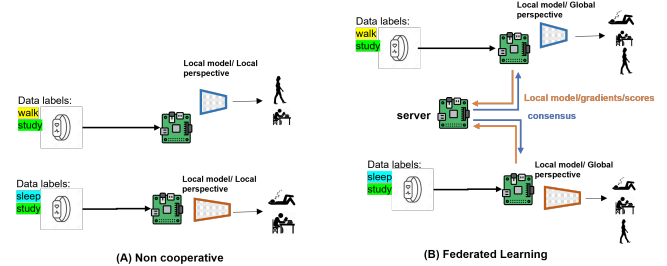


Fig. 1. Federated learning for CSI-based HAR. Each home client has access only to a subset of activity classes. FL enables a client to recognise *activities it has never observed locally* by absorbing knowledge from peers via soft-label distillation.

this gap by organising clients into architecture tiers and simultaneously sharing weights within tiers and feature carriers across tiers, outperforming both pure weight-sharing and pure KD baselines in semi-heterogeneous settings.

## Contributions.

- **HomeSenseKD**: an open benchmark of 7 FL algorithms on 4 datasets (2 CSI, 2 image) under uniform and non-IID partitions.
- **Dual-head tiered model**: a shared backbone with decoupled classification and distillation heads that scales from small to large devices.
- **Phase 1**: comprehensive IID/non-IID comparison across all algorithms, revealing the failure mode of all algorithms on HomeHAR.
- **Phase 2**: heterogeneity-distribution ablation of FedMD on HomeOccupancy (results to be added).
- **Phase 3**: direct FedMKS vs. FedDF efficiency comparison reporting accuracy, server/client compute load, and communication overhead (results to be added).

The remainder of this paper is structured as follows. Section II surveys related work. Section III defines the system model. Section IV describes all algorithms and the shared model. Section V presents three-phase experimental design and results. Section VI concludes.

## II. RELATED WORK

### A. Wi-Fi Sensing

Wi-Fi CSI sensing has progressed from coarse-grained indoor localisation to fine-grained activity and gesture recognition. Ma et al. [1] provide a comprehensive survey of CSI-based sensing techniques, cataloguing feature extraction strategies from amplitude and phase of subcarrier signals. Yousefi et al. [2] demonstrate that statistical features of CSI effectively distinguish human activities without device-attached sensors. Adib and Katabi [3] pioneered through-wall motion detection using commodity 802.11 hardware, establishing that reflections of Wi-Fi signals encode human presence. Wang et al. [4] introduced E-Eyes, one of the first systems mapping CSI amplitude patterns to specific indoor activities such as walking and falling. Chen et al. [5] showed that STFT-based time-frequency features of CSI enable real-time gesture classification with high accuracy. Han et al. [6] applied deep learning to CSI streams for healthcare HAR, achieving strong accuracy on activities of daily living with convolutional architectures. Jiang et al. [7] addressed environment dependence in CSI-based HAR, showing that domain-adaptation techniques improve cross-location generalisability. Liu et al. [8] proposed multi-task deep learning to jointly perform localisation and activity recognition from a single CSI stream.

### B. Federated Learning for Sensing

Sozinov et al. [9] were among the first to apply federated machine learning to HAR, demonstrating that local models collaboratively achieve accuracy competitive with a centralised model while preserving privacy. Wu et al. [10] extended FL to CSI-based sensing, showing that the non-IID nature of per-home activity distributions degrades FedAvg and motivating personalised federated approaches. Ek et al. [11] applied FL to WiFi-based user authentication, demonstrating that even simple weight-averaging benefits from multi-device participation when data distributions are complementary. Ouyang et al. [12] proposed ClusterFL, which groups FL clients by activity-distribution similarity, reducing inter-client disagreement and improving convergence on heterogeneous sensing datasets. Presotto et al. [13] introduced semantic-aware aggregation in FL for HAR, weighting client contributions by model confidence rather than dataset size. Zhao et al. [14] showed that FL accuracy degrades sharply when data are non-IID, especially for classification tasks, motivating specialised aggregation and distribution strategies.

### C. Federated Learning with Knowledge Distillation

McMahan et al. [15] introduced FedAvg, establishing the standard of aggregating locally trained weights at a central server and demonstrating strong performance on IID splits. Li et al. [16] extended FedAvg with a proximal regularisation term (FedProx) that stabilises training under system and statistical heterogeneity. Li and Wang [17] proposed FedMD, in which fully heterogeneous clients share only soft predictions on a public dataset, eliminating any constraint on client architecture at the cost of weaker gradient signal.

TABLE I  
SYMBOL TABLE.

| Symbol                     | Meaning                                                                                                         |
|----------------------------|-----------------------------------------------------------------------------------------------------------------|
| $K$                        | Number of clients ( $K=20$ )                                                                                    |
| $\mathcal{K}$              | Client index set $\{1, \dots, K\}$                                                                              |
| $R$                        | Number of FL communication rounds ( $R=50$ )                                                                    |
| $r$                        | Round index, $r \in \{1, \dots, R\}$                                                                            |
| $\mathcal{D}_k$            | Private labelled dataset of client $k$                                                                          |
| $n_k$                      | $ \mathcal{D}_k $ ; $n = \sum_k n_k$                                                                            |
| $\mathcal{Y}$              | Global label space; $C= \mathcal{Y} $                                                                           |
| $\mathcal{Y}_k$            | Local label subset, $\mathcal{Y}_k \subseteq \mathcal{Y}$                                                       |
| $\mathcal{D}^{\text{pub}}$ | Shared public (unlabelled) dataset                                                                              |
| $t_k$                      | Tier of client $k$ : S, M, or L                                                                                 |
| $\theta_k$                 | All trainable parameters of model $f_k$                                                                         |
| $\theta_k^{\text{bb}}$     | Backbone parameters (shared by classification and distillation heads)                                           |
| $\mathbf{W}_t^{(r)}$       | Server-side averaged classifier weights for tier $t$ at round $r$                                               |
| $d$                        | Distillation-head output dimension (feature carrier)                                                            |
| $s_k^{(r)}$                | Feature carrier from client $k$ at round $r$ , $s_k^{(r)} \in \mathbb{R}^{ \mathcal{D}^{\text{pub}}  \times d}$ |
| $\bar{s}^{(r)}$            | Aggregated global carrier at round $r$                                                                          |
| $T, B$                     | CSI window length (1500) and STFT bins (8)                                                                      |

Hinton et al. [18] established the foundational knowledge distillation framework—training a student to match teacher soft outputs—on which FedMD, FedAKD, and FedMKS build. Shen et al. [19] proposed federated mutual learning, enabling peer-to-peer distillation between clients without a public dataset. Lin et al. [20] proposed **FedDF**, in which the server collects client model outputs on a shared public dataset, constructs an ensemble prediction, and trains a fused student model via server-side knowledge distillation, improving accuracy and robustness compared to naive averaging. Unlike FedDF, which concentrates all distillation on the server, FedMKS distributes distillation across clients (Table III), substantially reducing server computation at the cost of higher client workload and communication overhead.

## III. SYSTEM MODEL

### A. Notation

### B. Network and Data Model

Consider a star-topology FL network with a central server and  $K=20$  clients. Each client  $k \in \mathcal{K}$  holds a private labelled dataset  $\mathcal{D}_k = \{(x_i^k, y_i^k)\}_{i=1}^{n_k}$ , where  $x_i^k \in \mathbb{R}^{T \times (4+B)}$  is a preprocessed CSI window and  $y_i^k \in \mathcal{Y}_k \subseteq \mathcal{Y}$  is the activity label. Under *IID* partitioning, each client receives  $n_c=30$  labelled examples per class. Under *non-IID* partitioning, class proportions are drawn from a Dirichlet distribution  $\mathbf{p}_k \sim \text{Dir}(\alpha)$  with  $\alpha=0.5$ , creating highly skewed per-client class distributions.

All clients share access to a freely available *public* dataset  $\mathcal{D}^{\text{pub}}$  (e.g. HomeHAR for HomeOccupancy clients) that carries no private labels and is used exclusively for knowledge transfer.

### C. Heterogeneous Model Architecture

Each client is assigned a compute tier  $t_k \in \{\text{S}, \text{M}, \text{L}\}$  reflecting its hardware budget. Every tier instantiates a *dual-head* model  $f_k = (\phi_k, h_k^{\text{clf}}, h_k^{\text{dist}})$ , where  $\phi_k$  is a shared

backbone parameterised by  $\theta_k^{\text{bb}}$  and the two heads are trained with decoupled losses:

- **Classification head**  $h_k^{\text{clf}}$ : trained on private data with cross-entropy loss  $\mathcal{L}_{\text{CE}}(\theta_k) = -\frac{1}{n_k} \sum_i \log p_k(y_i^k | x_i^k)$ .
- **Distillation head**  $h_k^{\text{dist}}$ : produces a  $d$ -dimensional feature carrier  $s_k^{(r)} = h_k^{\text{dist}}(\phi_k(\mathcal{D}^{\text{pub}})) \in \mathbb{R}^{|\mathcal{D}^{\text{pub}}| \times d}$ , aligned to the global aggregate via  $\mathcal{L}_{\text{KD}}(\theta_k) = \frac{1}{|\mathcal{D}^{\text{pub}}|} \|s_k^{(r)} - \bar{s}^{(r-1)}\|_F^2$ .

Backbone depth grows with tier: **S** uses Conv1D $\times 3$ ; **M** adds BiLSTM(128); **L** further adds a Transformer attention block.

#### D. Traditional FL Objective and Its Limitations

Standard FL (FedAvg [15]) seeks a single global model  $\theta^*$  minimising the weighted empirical risk:

$$\theta^* = \arg \min_{\theta} \sum_{k=1}^K \frac{n_k}{n} \mathcal{L}_{\text{CE}}^k(\theta), \quad (1)$$

aggregating client updates each round as  $\theta^{(r)} \leftarrow \sum_k (n_k/n) \theta_k^{(r)}$ . This formulation requires *all clients to share the same architecture*, making it inapplicable when  $t_j \neq t_k$ . FedProx [16] adds a proximal term  $\frac{\mu}{2} \|\theta_k - \theta^{(r-1)}\|^2$  to each client's loss to bound client drift under heterogeneous data, but still enforces architectural homogeneity.

#### E. Knowledge-Distillation-Based FL Objective

To support fully heterogeneous architectures, FedMD [17] replaces weight aggregation with shared-representation aggregation. The server maintains a global carrier matrix  $\bar{s}^{(r)} = \sum_k (n_k/n) s_k^{(r)}$  and each client minimises the combined objective:

$$\mathcal{L}_k(\theta_k) = \underbrace{\mathcal{L}_{\text{CE}}(\theta_k; \mathcal{D}_k)}_{\text{private classification}} + \lambda \underbrace{\mathcal{L}_{\text{KD}}(\theta_k; \mathcal{D}^{\text{pub}}, \bar{s}^{(r-1)})}_{\text{KD alignment}}, \quad (2)$$

where  $\lambda$  balances local accuracy with global knowledge absorption. Because the carrier  $\bar{s}^{(r)}$  lives in a *shared  $d$ -dimensional feature space*, clients with entirely different backbone architectures can participate in the same aggregation round.

#### F. Problem Definition

**Goal.** Given  $K$  clients with heterogeneous private datasets  $\{\mathcal{D}_k\}$  and heterogeneous model tiers  $\{t_k\}$ , find per-client classifiers  $\{f_k^{\text{clf}}\}$  that jointly maximise mean validation accuracy across clients, subject to three constraints:

- 1) *Privacy*:  $\mathcal{D}_k$  never leaves client  $k$ ; only  $s_k^{(r)}$  and/or  $\theta_k^{\text{bb}}$  are transmitted.
- 2) *Architectural heterogeneity*: no direct weight transfer between clients of different tiers ( $t_j \neq t_k$ ).
- 3) *Communication bound*: per-round client payload is the sum of a within-tier weight term and a cross-tier knowledge term. Pure weight-sharing methods transfer  $\mathcal{O}(|\theta|)$ , pure KD methods transfer  $\mathcal{O}(|\mathcal{D}^{\text{pub}}| \cdot d)$ , and FedMKS transfers both within-tier classifier weights  $\mathcal{O}(|\theta_k|)$  and cross-tier feature carriers  $\mathcal{O}(|\mathcal{D}^{\text{pub}}| \cdot d)$ , but never full weights across tiers. Every client receives its

tier's aggregated weights *and* the global carrier; it uses the carrier as targets to train on  $\mathcal{D}^{\text{pub}}$ .

FedMKS (Section IV) addresses all three constraints simultaneously by combining within-tier weight averaging (Eq. 1 applied per tier) with cross-tier feature-carrier distillation (Eq. 2), bridging the gap between FedAvg efficiency and FedMD flexibility.

## IV. METHODS

### A. Tiered Dual-Head Architecture

All three tiers share the following backbone structure: **Small**  $\rightarrow$  Conv1D(64,5)  $\rightarrow$  Conv1D(128,3)  $\rightarrow$  Conv1D(256,3); **Medium**  $\rightarrow$  Small + BiLSTM(128); **Large**  $\rightarrow$  Medium + Transformer block (MHA-4-heads, FFN, LayerNorm, GAP). The *classification head* (Dense  $\rightarrow$  Softmax, CE loss) is trained on  $\mathcal{D}_k$ ; the *distillation head* (Dense  $\rightarrow$  ReLU with  $d=64$  output units, MSE loss on  $\bar{s}$ ) is trained on  $\mathcal{D}^{\text{pub}}$ . Gradient flows are fully decoupled between heads; only the backbone is shared.

### B. FedMKS: Mixed Knowledge Sharing

FedMKS (Fig. 2, Fig. 3) is the proposed algorithm that addresses the limitations of both FedAvg and FedMD in heterogeneous networks. FedAvg requires all clients to share a single architecture, so it must use the smallest model in a heterogeneous network—sacrificing accuracy for capable clients. FedMD shares only feature carriers, accommodating full heterogeneity, but misses the accuracy gains of direct weight sharing. FedMKS simultaneously: (i) shares weights within clients of the same tier (intra-tier FedAvg); (ii) distills knowledge across all tiers via aggregated feature carriers on the public set; and (iii) applies Mixup augmentation to the public set before distillation (following FedAKD).

In round  $r=1$  the server holds no global state, so each client trains only on its private data and then uploads its initial feature carrier and classifier weights. From round  $r=2$  onward, the server sends every client its tier's averaged classifier weights  $\mathbf{W}_{t_k}^{(r-1)}$  and the global carrier  $\bar{s}^{(r-1)}$ ; the client loads the weights, fine-tunes the distillation head on the public set using  $\bar{s}^{(r-1)}$  as targets, and then trains the classification head on its private data for  $E_{\text{loc}}$  epochs. It returns the new carrier  $s_k^{(r)}$  and classifier weights  $w_k^{(r)}$  to the server.

Algorithm 1 formalises one FedMKS round.

**Comparison with FedDF.** FedDF [20] addresses the same heterogeneous FL problem via server-side ensemble distillation: the server runs all client models on  $\mathcal{D}^{\text{pub}}$ , forms an ensemble  $\bar{s}^{(r)} = \sum_k \beta_k f(\mathcal{D}^{\text{pub}}, \theta_k)$ , and trains a fused student by minimising  $D_{\text{KL}}(\bar{s}_\tau \| f_\tau(\mathcal{D}^{\text{pub}}, \theta))$ . This centralises both inference and gradient computation on the server, incurring  $\mathcal{O}(KPF + E_{\text{kd}}P(\mathcal{F}+\mathcal{B}))$  operations per round, where  $P=|\mathcal{D}^{\text{pub}}|$  and  $\mathcal{F}, \mathcal{B}$  denote forward- and backward-pass costs. FedMKS inverts this design: the server performs only tensor aggregation  $\mathcal{O}(\sum_t |\mathbf{W}_t| + KPd)$  with no neural-network passes, shifting the distillation burden to clients—making it far more suitable when the coordinator is a lightweight edge gateway (Table III).

Table II summarises all evaluated algorithms.

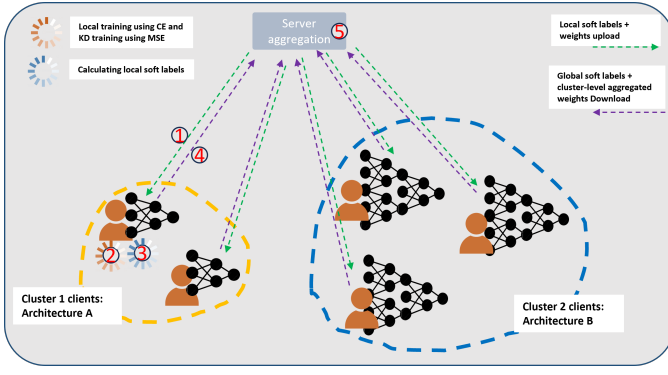


Fig. 2. **FedMKS overview.** In a heterogeneous network (Small, Medium, Large clients), FedAvg must impose a single small architecture; FedMD shares only carriers but loses weight-sharing benefits. FedMKS partitions clients into tier clusters, applying intra-tier weight aggregation alongside cross-tier feature-carrier distillation, achieving the best of both paradigms.

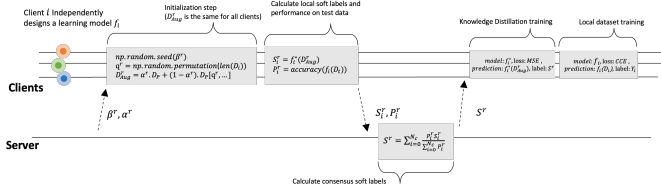


Fig. 3. **FedMKS round timeline.** From round  $r=2$  onward, every client receives the global carrier  $\bar{s}$  and its tier’s averaged classifier weights  $\mathbf{W}_{t_k}$  from the server; runs  $E_{kd}$  KD epochs on the public set using  $\bar{s}$  as targets; runs  $E_{loc}$  local epochs on private data; then returns its updated carrier  $s_k$  and classifier weights  $w_k$  to the server. Round  $r=1$  bootstraps by local training only, producing the first carriers and weights.

## V. EXPERIMENTS AND RESULTS

### A. Experiment Design

**Datasets.** *HomeOccupancy* [21]: 3-class CSI (empty/sleep/work), sessions 1–2 as train, session 3 as test; public data is HomeHAR. *HomeHAR* [21]: 7-class CSI (drink/eat/empty/sleep/ smoke/watch/work), all sessions combined with 80/20 split; public data is HomeOccupancy. *MNIST*: 10-class digit images; public data is CIFAR-10. *CIFAR-10*: 10-class object images; public data is MNIST.

**Hyperparameters** (Table IV):  $R=50$  rounds,  $K=20$  clients,  $n_c=30$  samples/class/client (IID),  $\alpha=0.5$  (Dirichlet non-IID),  $E_{loc}=4$ ,  $E_{kd}=2$ ,  $\mu=0.01$  (FedProx),  $n_{align}=10,000$ ,  $T=1500$ ,  $B=8$ . Tier distribution (Phase 1): 35% S, 35% M, 30% L. All results are mean validation accuracy across 3 independent seeds.

**Phase 1** evaluates all 7 algorithms on 4 datasets under uniform tier heterogeneity (35% S/35% M/30% L) for both IID and non-IID data partitions. **Phase 2** ablates tier-heterogeneity distribution on HomeOccupancy for FedMD across three configurations: *all\_small* (100% S), *uniform* (35/35/30), and *skewed* (60% S/30% M/10% L). **Phase 3** directly compares FedMKS and FedDF on HomeOccupancy and HomeHAR under both IID and non-IID partitions (uniform tier distribution), reporting (i) validation accuracy, (ii) per-round server computation time, (iii) per-round mean client

### Algorithm 1 FedMKS “one communication round

**Require:** Clients  $\mathcal{K}$ ; tiers  $t \in \{S, M, L\}$ ;  $\mathcal{D}^{pub}$ ;  $E_{kd}$ ,  $E_{loc}$ ; round  $r$ ; current global carrier  $\bar{s}^{(r-1)}$  and tier weights  $\{\mathbf{W}_t^{(r-1)}\}$   
**Ensure:** Updated global carrier  $\bar{s}^{(r)}$  and tier weights  $\{\mathbf{W}_t^{(r)}\}$

- 1: **if**  $r = 1$  **then**
- 2:   **Server:** send empty payload to each  $k$  ▷ bootstrap
- 3: **else**
- 4:   **Server:** broadcast  $\bar{s}^{(r-1)}$  and  $\mathbf{W}_t^{(r-1)}$  to each  $k$
- 5: **end if**
- 6: **for each client**  $k$  **in parallel do**
- 7:   **if**  $r > 1$  and  $\mathbf{W}_{t_k}^{(r-1)}$  received **then**
- 8:     Load classifier weights  $\leftarrow \mathbf{W}_{t_k}^{(r-1)}$
- 9:     Receive carrier  $\bar{s}^{(r-1)}$  into public-set targets
- 10:    Fine-tune dist. head on  $\mathcal{D}^{pub}$  for  $E_{kd}$  epochs minimising  $\mathcal{L}_{MSE}(f_k^{dist}(\cdot), \bar{s}^{(r-1)})$
- 11:    **end if**
- 12:    Train clf. head on private data  $\mathcal{D}_k$  for  $E_{loc}$  epochs minimising  $\mathcal{L}_{CE}$
- 13:    Return  $s_k^{(r)} \leftarrow f_k^{dist}(\mathcal{D}^{pub})$  and  $w_k^{(r)} \leftarrow \theta_k$
- 14: **end for**
- 15: **Server:**  $\bar{s}^{(r)} \leftarrow \sum_k (n_k/n) s_k^{(r)}$
- 16: **Server:**  $\mathbf{W}_t^{(r)} \leftarrow \text{FedAvg}(\{w_k^{(r)} : t_k=t\})$  for each tier  $t$

TABLE II  
ALGORITHM SUMMARY (✓ = FEATURE USED; † PHASE 3 ONLY).

| Algo.   | Soft | Wts | Topology      |
|---------|------|-----|---------------|
| FedMD   | ✓    | –   | Heterogeneous |
| FedAKD  | ✓    | –   | Heterogeneous |
| FedMKS  | ✓    | ✓   | Semi-het.     |
| FedDF†  | ✓    | ✓   | Heterogeneous |
| FedAvg  | –    | ✓   | Homogeneous   |
| FedProx | –    | ✓   | Homogeneous   |
| Local   | –    | –   | Isolated      |
| Central | –    | –   | Centralised   |

computation time, and (iv) total communication bytes per round.

### B. Results

1) *Phase 1 “Algorithm Comparison”*: Fig. 4 shows validation accuracy curves for all 8 experiment conditions (4 datasets  $\times$  2 data partitions) over 50 federated rounds.

**HomeOccupancy.** Under IID, FedMKS (90.3%) and FedAvg (90.0%) closely approach the centralised ceiling (92.1%), while FedMD (86.3%) and FedAKD (87.7%) lag due to the absence of weight sharing. Local training plateaus at 81.3%, confirming the value of collaboration. Under non-IID, centralised performance remains stable (92.1%), but federated methods diverge sharply: FedMD improves slowly to 63.4% while weight-sharing methods stagnate below 47%. Local training (60.0%) outperforms weight-sharing under non-IID, a well-known phenomenon when Dirichlet skew prevents effective aggregation.

**HomeHAR (failure mode).** All algorithms converge to  $\approx 31\%$  accuracy on HomeHAR IID— essentially the majority-class fraction of the 7-class dataset. This applies even to the centralised model, confirming that the failure is at the

TABLE III  
FEDMKS VS. FEDDF: ALGORITHMIC COMPARISON USING THE NOTATION OF SECTION III.  $P \triangleq |\mathcal{D}^{\text{pub}}|$  (PUBLIC SAMPLES);  $\mathcal{F}$ ,  $\mathcal{B}$ : FORWARD- AND BACKWARD-PASS COSTS (DISTINCT FROM CSI WINDOW  $T$  AND STFT BINS  $B$ ).

| Aspect                | FedMKS (proposed)                                                                                                                                                    | FedDF [20]                                                                                                                                 |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Objective             | Intra-tier weight sharing combined with cross-tier feature-carrier distillation on $\mathcal{D}^{\text{pub}}$ .                                                      | Server-side ensemble distillation: a fused model is trained to match the ensemble of client outputs on $\mathcal{D}^{\text{pub}}$ .        |
| Weight aggregation    | Per-tier FedAvg: $\mathbf{W}_t^{(r)} = \frac{\sum_{k: t_k=t} n_k \theta_k}{\sum_{k: t_k=t} n_k}$ .                                                                   | Compatible client models may be averaged first; not the primary fusion mechanism.                                                          |
| Knowledge aggregation | Clients upload $s_k^{(r)}$ ; server computes $\bar{s}^{(r)} = \sum_k \frac{n_k}{n} s_k^{(r)}$ ; no server-side inference.                                            | Server runs all client models on $\mathcal{D}^{\text{pub}}$ : $\bar{s}^{(r)} = \sum_k \beta_k f(\mathcal{D}^{\text{pub}}; \theta_k)$ .     |
| Distillation          | Each client $k$ minimises $\mathcal{L}_k = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{KD}}(\theta_k; \bar{s}^{(r-1)})$ locally for $E_{\text{kd}}$ epochs. | Server trains a fused model: $\theta^* = \arg \min_{\theta} D_{\text{KL}}(\bar{s}_{\tau} \  f_{\tau}(\mathcal{D}^{\text{pub}}; \theta))$ . |
| KD location           | Clients, in parallel.                                                                                                                                                | Central server only.                                                                                                                       |
| Server workload       | Tensor aggregation only: $O(\sum_t  \mathbf{W}_t  + KPd)$ ; no forward or backward passes.                                                                           | Client inference + student optimisation: $O(KPF + E_{\text{kd}}P(\mathcal{F} + \mathcal{B}))$ .                                            |
| Communication         | Per client: $ \theta_k $ weights + $P \times d$ feature-carrier matrix.                                                                                              | Per client: $ \theta_k $ weights; soft labels remain on the server.                                                                        |
| Main trade-off        | Lower server compute; higher client compute and communication overhead.                                                                                              | Lower client KD workload; higher server compute and memory.                                                                                |
| Suitable setting      | Resource-constrained IoT or edge coordinator.                                                                                                                        | Computationally capable central server.                                                                                                    |

TABLE IV  
HYPERPARAMETER SETTINGS (ALL EXPERIMENTS).

| Parameter                     | Value  |
|-------------------------------|--------|
| Rounds $R$                    | 50     |
| Clients $K$                   | 20     |
| IID samples / class / client  | 30     |
| Non-IID Dirichlet $\alpha$    | 0.5    |
| Local epochs $E_{\text{loc}}$ | 4      |
| KD epochs $E_{\text{kd}}$     | 2      |
| Carrier dimension $d$         | 64     |
| FedProx $\mu$                 | 0.01   |
| Alignment samples             | 10 000 |
| CSI window $T$                | 1500   |
| STFT bins $B$                 | 8      |

dataset level, not algorithmic. HomeHAR presents severe data scarcity: 30 samples per class per client (IID) and fewer under non-IID yields  $<300$  training windows in the most constrained case. With 7 classes and short CSI recordings, the per-window feature extractor lacks sufficient diversity to learn discriminative patterns. Training a centralised Transformer on the full HomeHAR corpus (via `dl.py --dataset home_har`) is required to establish a meaningful upper bound.

**MNIST.** MNIST is an easier benchmark that confirms algorithm ordering under ideal conditions. IID: Central 99.0%, FedMKS 97.6%, FedAvg/FedProx 97.1%/97.2%, FedMD/FedAKD/Local all  $\approx 95.8\%$ . Non-IID: FedAvg 97.7%, FedMKS 97.0%, FedProx 97.5% remain competitive, while FedMD/FedAKD/Local plateau at  $\approx 85\%$ , confirming that soft-label-only methods fail under distribution shift.

**CIFAR-10.** CIFAR-10 reveals the fundamental difficulty

of cross-dataset public/private knowledge transfer (public is MNIST, private is CIFAR-10). All FL methods fall 10–30% below centralised. FedAvg (53.4%) marginally leads FedMKS (52.0%) under IID; under non-IID FedProx (47.7%) edges out FedAvg (47.1%) and FedMKS (46.0%). KD-only methods (FedMD 40.7%, FedAKD 40.9%) barely exceed local training (40.6%), suggesting MNIST soft labels provide negligible signal for CIFAR-10 learning.

2) *Phase 2  $\Delta \epsilon$ ” Heterogeneity Ablation:* Phase 2 investigates how tier-heterogeneity distribution affects FedMD on HomeOccupancy, comparing `all_small`, `uniform`, and `skewed` configurations under both IID and non-IID settings. *Experiments are ongoing; results will be added upon completion.* Fig. 5 is a placeholder.

3) *Phase 3 — FedMKS vs. FedDF:* Phase 3 provides a direct empirical comparison of FedMKS and FedDF on HomeOccupancy and HomeHAR under both IID and non-IID settings (uniform tier distribution,  $R=50$  rounds). Beyond validation accuracy, the evaluation records (ii) per-round server computation time (aggregation only for FedMKS vs. client inference and student optimisation for FedDF), (iii) per-round mean client computation time, and (iv) total communication bytes per round, enabling the complexity analysis of Table III to be validated experimentally. *Experiments are ongoing; results will be added upon completion.* Fig. 6 is a placeholder.

## VI. CONCLUSION

HomeSenseKD presents a systematic benchmark for heterogeneous federated knowledge distillation applied to CSI-based indoor activity sensing. The dual-head tiered model and the FedMKS algorithm demonstrate that combining intra-tier weight sharing with cross-tier feature-carrier distilla-

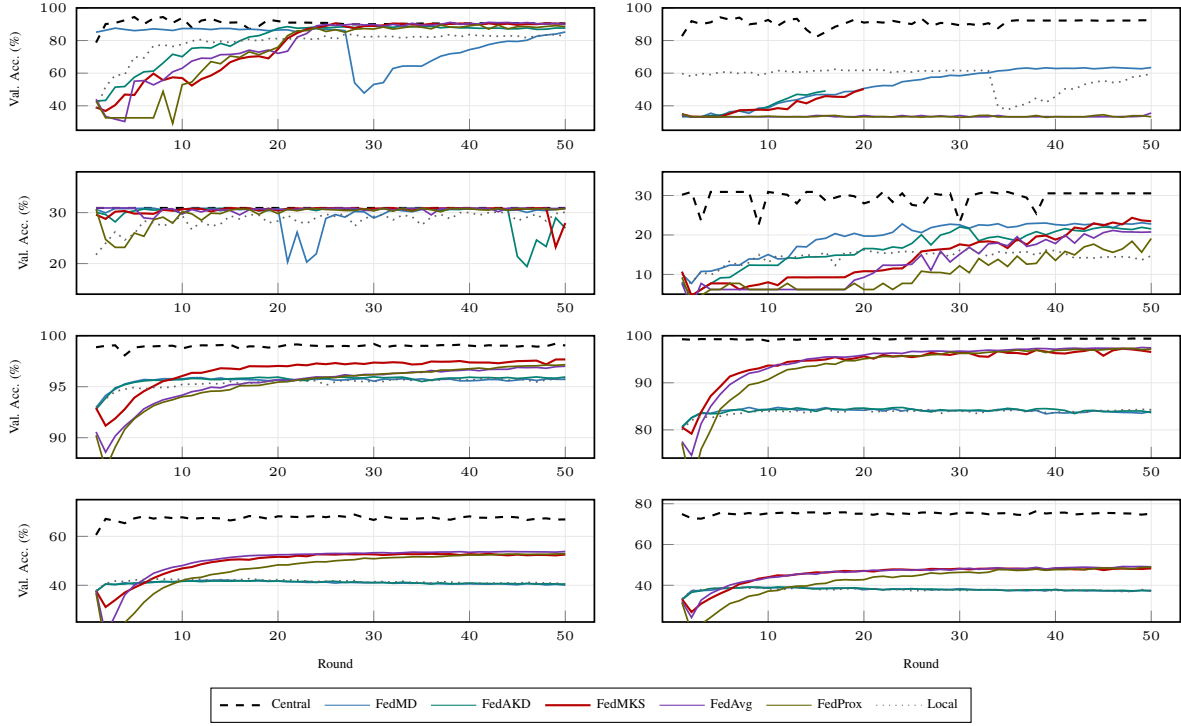


Fig. 4. **Phase 1: Validation accuracy over 50 federated rounds** (mean across 3 seeds). Left column: IID; right column: non-IID (Dirichlet  $\alpha=0.5$ ). Rows: HomeOccupancy, HomeHAR, MNIST, CIFAR-10. All algorithms use uniform tier distribution (35% S/35% M/30% L). HomeHAR curves collapse to the majority-class baseline ( $\approx 31\%$ ), revealing a dataset-level challenge independent of algorithm choice. On HomeOccupancy IID, FedMKS and FedAvg match the centralised ceiling; under non-IID, weight-sharing methods degrade sharply while Local and FedMD remain more robust. On MNIST, all FL methods approach centralised accuracy under IID; non-IID separates FedMKS/FedAvg/FedProx from FedMD/FedAKD/Local. On CIFAR-10, the task difficulty suppresses all federated methods by  $> 10\%$  relative to Central, underscoring the limits of knowledge distillation with small per-client datasets. † FedAKD and FedMKS curves on HomeOccupancy non-IID terminated early (16 and 20 rounds respectively) due to training instability.

Phase 2 Results  
(to be added after experiment completion)

Fig. 5. **Phase 2 placeholder.** Validation accuracy of FedMD under three tier-heterogeneity configurations (`all_small`, `uniform`, `skewed`) on HomeOccupancy IID and non-IID. Results to be added.

tion achieves near-centralised accuracy on HomeOccupancy ( $\Delta=1.8\%$  under IID) while preserving full architectural heterogeneity. Phase 1 results on four datasets expose a clear hierarchy: weight-sharing methods dominate under IID; soft-label methods are more robust under non-IID when distributions are highly skewed; and HomeHAR reveals that data scarcity, not algorithm design, is the binding constraint for challenging 7-class CSI recognition. Phase 3 will validate the theoretical efficiency advantage of FedMKS over FedDF by measuring server compute, client compute, and communication overhead directly, providing empirical grounding for the analysis in Table III. The public benchmark, code, and datasets provide a reproducible foundation for future work on privacy-preserving

Phase 3 Results  
(FedMKS vs. FedDF: accuracy, server/client  
compute, and communication overhead per round;  
to be added after experiment completion)

Fig. 6. **Phase 3 placeholder.** Validation accuracy and efficiency metrics (server compute, client compute, communication bytes per round) for FedMKS vs. FedDF on HomeOccupancy and HomeHAR under IID and non-IID data partitions. Results to be added.

sensing.

## VII. LIMITATIONS AND FUTURE WORK

**HomeHAR data scarcity.** The 7-class HomeHAR dataset yields insufficient windows for reliable multi-class federated learning under 30-sample-per-class constraints. Future work should collect additional recording sessions and explore data augmentation (Mixup, time-warping) or self-supervised pre-training to improve sample efficiency.

**Single tier distribution in Phase 1.** Phase 1 uses only uniform tier distribution (35/35/30). Varying heterogeneity as

in Phase 2 would reveal whether FedMKS advantages are consistent across tier compositions.

**Communication cost.** FedMKS transmits both soft labels and per-tier weights, incurring higher communication overhead than FedMD. Compression (quantisation, top- $k$  sparsification) of weight updates would improve deployment feasibility on bandwidth-constrained IoT networks.

**Cross-domain public datasets.** CIFAR-10 experiments show that a public dataset from a different domain (MNIST) provides little useful signal. Investigating domain-adapted public data, or constructing virtual public datasets from client gradients, remains an open problem.

**Asynchronous FL.** HomeSenseKD assumes synchronous rounds. Real deployments exhibit client dropouts and straggler effects; adapting FedMKS to asynchronous aggregation is an important practical extension.

#### CODE AND DATA AVAILABILITY

Source code is available at <https://github.com/gadm21/HomeSenseKD>. HomeOccupancy and HomeHAR datasets are hosted on HuggingFace at <https://huggingface.co/gadgadgad>.

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